Online Learning — Lecture Notes, November 6, 2026

Review, learning with covariates, and an introduction to FTRL

1 Review: sequential learning and regret

1.1 Time series and sampling frequency

A time series is an observed realization of a stochastic process $(Y_t)_{t\in\mathcal{T}}$. The index t represents the observation time or period. Its **sampling frequency** specifies how often the variable is measured: hourly temperature, daily stock prices, annual student enrolment, or monthly call volume.

A time index does not necessarily imply dependence, but it allows us to describe mechanisms in which the present depends on the past. In Online Learning, decisions are made sequentially, without requiring observations to be independent and identically distributed.

Definition 1.1 (Convex decision set)**.** A set $\mathcal{K} \subseteq \mathbb{R}^d$ is **convex** if

$$\forall u, v \in \mathcal{K},\ \forall \tau \in [0,1], \qquad \tau u + (1-\tau)v \in \mathcal{K}.$$

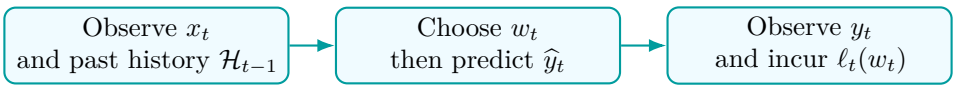
Thus, a convex combination of two feasible decisions is still in $\mathcal{K}$. An interval in $\mathbb{R}$ and a Euclidean ball in $\mathbb{R}^d$ are convex.

1.2 The online protocol: a strict chronology

Let $\mathcal{K}$ be the set of admissible parameters. At time t, the learner chooses $w_t \in \mathcal{K}$ using only information available *before* the target y_t is revealed. The learner then incurs a loss $\ell_t(w_t)$ and updates the decision. Formally, if $\mathcal{H}_t$ denotes the history revealed up to time t, we write

$$w_{t+1} = \mathcal{A}_t(\mathcal{H}_t), \qquad \mathcal{H}_t = \{(x_s, y_s) : 1 \le s \le t\}.$$

The covariate x_t may be available before the decision, for example when a promotion is announced in advance. We therefore allow w_t to depend on x_t *when the protocol permits it*, but **never** on y_t.



Definition 1.2 (Fixed comparator and regret)**.** The best *fixed parameter in hindsight* is

$$w^\star \in \arg\min_{u\in\mathcal{K}} \sum_{t=1}^{T} \ell_t(u).$$

Regret against a fixed decision u and standard regret are

$$\mathrm{Reg}_T(u) = \sum_{t=1}^{T} [\ell_t(w_t) - \ell_t(u)],$$

$$\mathrm{Reg}_T = \sum_{t=1}^{T} \ell_t(w_t) - \min_{u\in\mathcal{K}} \sum_{t=1}^{T} \ell_t(u).$$

Average regret is Reg_T/T. An algorithm is said to have **no regret** if $\mathrm{Reg}_T/T \to 0$ as $T \to \infty$.

Regret measures the difference in cumulative loss relative to a constant decision that knows the entire sequence. It does not directly measure absolute forecasting error. In general, realized regret can be negative; a sublinear upper bound therefore first guarantees $\limsup_T \mathrm{Reg}_T/T \le 0$. For FTL with exact minimizers, standard regret is nonnegative, and such a bound indeed implies $\mathrm{Reg}_T/T \to 0$.

1.3 Subgradients and optimality conditions

Definition 1.3 (Subgradient)**.** For a convex function $f : \mathbb{R}^d \to \mathbb{R}$, a vector $g \in \mathbb{R}^d$ is a subgradient at u if

$$f(v) \geq f(u) + \langle g, v - u \rangle, \qquad \forall v \in \mathbb{R}^d.$$

The set of these vectors is the **subdifferential** $\partial f(u)$. If f is differentiable at u, then $\partial f(u) = \{\nabla f(u)\}$.

Geometrically, $v \mapsto f(u) + \langle g, v - u \rangle$ is a global affine lower bound on f. In particular,

$$u^\star \in \arg\min_{u \in \mathbb{R}^d} f(u) \quad \Longleftrightarrow \quad 0 \in \partial f(u^\star).$$

This condition completely characterizes *unconstrained* optimality for a convex function. Over a constrained set $\mathcal{K}$, one instead considers $f + \iota_{\mathcal{K}}$ or uses an optimality condition adapted to the constraints.

Example 1.4 (Absolute loss)**.** For $f(u) = |u|$, the left and right derivatives at zero are -1 and 1, respectively: the function is not differentiable there. However,

$$\partial |u| = \begin{cases} \{-1\}, & u < 0, \\ [-1, 1], & u = 0, \\ \{1\}, & u > 0. \end{cases}$$

Since $0 \in \partial |0|$, the point $u^\star = 0$ is indeed a minimizer.

1.4 Follow the Leader and regret assumptions

Definition 1.5 (Follow the Leader (FTL))**.** Starting from an initial choice $w_1 \in \mathcal{K}$, FTL defines

$$w_{t+1} \in \arg\min_{u \in \mathcal{K}} \sum_{s=1}^{t} \ell_s(u).$$

The update w_{t+1} uses the loss at time t; the decision w_t, taken at the beginning of the round, does not yet use it.

Theorem 1.6 (FTL: logarithmic regret bound)**.** *If $\mathcal{K}$ is convex, exact FTL minimizers exist, every ℓ_t is α-strongly convex ($\alpha > 0$), and there exist $g_t \in \partial \ell_t(w_t)$ with $\|g_t\|_2 \leq G$, then for every $u \in \mathcal{K}$,*

$$\mathrm{Reg}_T(u) \leq \frac{G^2}{2\alpha} \sum_{t=1}^{T} \frac{1}{t} \leq \frac{G^2}{2\alpha}(1 + \log T).$$

Since $\log T/T \to 0$, the standard regret of FTL is then asymptotically negligible on average. **These assumptions are not automatic.** The scalar loss $\ell_t(\theta) = \frac{1}{2}(y_t - \theta)^2$ is 1-strongly convex; its gradient $\theta - y_t$ is bounded only if both decisions and observations are bounded.

> **Important note.** As soon as the forecast is enriched with several parameters, the squared loss for a single observation is generally *not* strongly convex over the entire parameter space. The above theorem therefore cannot be applied mechanically to the model with covariates introduced below.

1.5 A closer look at the assumptions: curvature and bounded gradients

The logarithmic regret theorem requires **two distinct properties**: strictly positive curvature of every loss and control of the subgradients at the decisions actually played.

Property 1.7 (Scalar squared loss). *For the loss $\ell_t(\theta) = \frac{1}{2}(\theta - y_t)^2$, we have*

$$\ell_t'(\theta) = \theta - y_t, \qquad \ell_t''(\theta) = 1.$$

Hence ℓ_t is α-strongly convex with $\alpha = 1$, regardless of y_t. Strong convexity does not, however, provide a bound on $|\ell_t'(\theta)|$.

On $\mathcal{K} = \mathbb{R}$. The gradient is not uniformly bounded: for fixed y_t, $|\theta - y_t| \to \infty$ as $|\theta| \to \infty$. The same problem occurs if $\mathcal{K}$ is bounded but y_t can take arbitrarily large values.

On $\mathcal{K} = [0, K]$. If we *also* assume $y_t \in [0, K]$ for every t, then for all $\theta \in \mathcal{K}$,

$$|\ell_t'(\theta)| = |\theta - y_t| \le K.$$

We may therefore choose $G = K$ and $\alpha = 1$. If the loss were defined as $(\theta - y_t)^2$ without the factor $\frac{1}{2}$, we would instead have $\alpha = 2$ and $G = 2K$.

> *Remark* 1.8 (Absolute loss). For $\ell_t(\theta) = |\theta - y_t|$, every subgradient lies in $[-1, 1]$, so $G = 1$ even on $\mathbb{R}$. But this loss is convex only, *not strongly convex*: the preceding logarithmic regret theorem does not apply.

2 Illustrative notebook: introducing a covariate

2.1 From constant forecasts to conditional predictions

In the original notebook, we predict daily demand y_t using a constant θ_t. The loss is

$$\ell_t^{\text{base}}(\theta) = \frac{1}{2}(y_t - \theta)^2, \qquad \theta_t \in \arg\min_{\theta \in \mathbb{R}} \sum_{s=1}^{t-1} \ell_s^{\text{base}}(\theta).$$

For $t \ge 2$, FTL therefore corresponds to the historical mean $\theta_t = (t-1)^{-1} \sum_{s<t} y_s$.

We now wish to exploit information that is available **before the sale**: is the store running a promotion?

$$x_t = \begin{cases} 1, & \text{promotion announced for day } t, \\ 0, & \text{no promotion on day } t. \end{cases}$$

The new forecast is

$$\widetilde{\theta}_t = \underbrace{\theta_t}_{\text{baseline demand}} + \underbrace{\beta_t x_t}_{\text{promotion adjustment}}.$$

Here θ_t is the intercept, whereas β_t is the *predictive* effect associated with a promotion. It must not be mistaken for a causal effect: in practice, promotions may be scheduled according to expected demand.

Definition 2.1 (Linear loss model with a covariate)**.** Define the parameter vector and feature vector as

$$w = \begin{pmatrix} \theta \\ \beta \end{pmatrix} \in \mathbb{R}^2, \qquad z_t = \begin{pmatrix} 1 \\ x_t \end{pmatrix} \in \mathbb{R}^2.$$

The model prediction is $\widehat{y}_t(w) = z_t^\top w = \theta + \beta x_t$. Its squared loss on day t is

$$\widetilde{\ell}_t(w) = \frac{1}{2}\left(y_t - z_t^\top w\right)^2 = \frac{1}{2}(y_t - \theta - \beta x_t)^2.$$

FTL **estimates both coefficients jointly**:

$$w_t \in \arg\min_{w \in \mathcal{K}} \frac{1}{2} \sum_{s=1}^{t-1} (y_s - z_s^\top w)^2, \quad t \geq 2.$$

The distinction matters: we do not first choose a historical average and then add an arbitrary coefficient. **Both parameters are re-estimated from past observations by solving the same optimization problem.**

2.2 Binary feature: an explicit FTL solution

To understand precisely what the notebook must compute, define

$$I_j(t) = \{s < t : x_s = j\}, \qquad n_j(t) = |I_j(t)|, \quad j \in \{0, 1\}.$$

When $n_0(t), n_1(t) > 0$, define the conditional historical means

$$\overline{y}_{j,t} = \frac{1}{n_j(t)} \sum_{s \in I_j(t)} y_s.$$

The FTL criterion then splits into two groups:

$$F_{t-1}(\theta, \beta) = \frac{1}{2} \sum_{s \in I_0(t)} (y_s - \theta)^2 + \frac{1}{2} \sum_{s \in I_1(t)} (y_s - \theta - \beta)^2.$$

Proposition 2.2 (FTL with a promotion indicator)**.** *If both categories have been observed at least once, the minimizer is unique and satisfies*

$$\theta_t = \overline{y}_{0,t}, \qquad \beta_t = \overline{y}_{1,t} - \overline{y}_{0,t}.$$

Consequently,

$$\widetilde{\theta}_t = \begin{cases} \overline{y}_{0,t}, & x_t = 0, \\ \overline{y}_{1,t}, & x_t = 1. \end{cases}$$

The enriched model thus computes a historical mean for each group*, depending on promotion status.*

Example 2.3 (Calculation on day 4)**.** Suppose we have observed $y_1 = 10$, $y_2 = 18$, $y_3 = 12$, with $x_1 = 0$, $x_2 = 1$, $x_3 = 0$. The mean without promotions is $(10 + 12)/2 = 11$, whereas the mean on promotion days is 18. Thus, before day 4, FTL yields

$$\theta_4 = 11, \qquad \beta_4 = 18 - 11 = 7.$$

If $x_4 = 1$, the enriched forecast is 18, compared with $40/3 \simeq 13.33$ for the model without covariates. If actual demand is $y_4 = 21$, their squared losses are, respectively, 4.5 and $529/18 \simeq 29.39$. This is an illustrative example, not a guaranteed dominance result.

Proof. We obtain $\partial_\beta F_{t-1} = n_1(t)(\theta + \beta - \overline{y}_{1,t})$ and

$$\partial_\theta F_{t-1} = n_0(t)(\theta - \overline{y}_{0,t}) + n_1(t)(\theta + \beta - \overline{y}_{1,t}).$$

Setting both derivatives to zero gives the two expressions. Since both groups are nonempty, the cumulative Hessian is positive definite: the minimizer is unique. □

Important note. Cold start. As long as either category $x = 0$ or $x = 1$ has not yet been observed, the coefficients θ and β cannot be separately identified by least squares. To define a deterministic policy in this exercise, we therefore set $\beta_t = 0$ and $\theta_t = \frac{1}{t-1}\sum_{s<t} y_s$ for $t \geq 2$. This choice remains an FTL minimizer of the historical loss. On the first round, set $\theta_1 = 10$ and $\beta_1 = 0$.

Remark 2.4 (Why the logarithmic theorem does not apply directly). For a single observation, the Hessian of the enriched loss is

$$\nabla^2 \widetilde{\ell}_t(w) = z_t z_t^\top = \begin{pmatrix} 1 & x_t \\ x_t & x_t^2 \end{pmatrix}.$$

This matrix has rank one and a zero eigenvalue: $\widetilde{\ell}_t$ is not α-strongly convex over $\mathbb{R}^2$ for any $\alpha > 0$. On the other hand, the *cumulative* Hessian becomes positive definite when both categories occur. By itself, this does not imply the theorem's uniform strong-convexity assumption on each individual loss.

2.3 Student work in the Python notebook

Objective. Starting from the illustrative notebook provided, implement an FTL strategy that uses a binary promotion covariate, then rigorously compare its forecasts and losses with those of the original FTL without covariates. **No future observation may enter a decision at time t.**

Exercise 2.1 (Contextual FTL: promotions and demand)**. Data.** Keep the time axis and random seed already used in the notebook. If it already contains a promotion indicator known before day t, use it. Otherwise, generate a sequence $x_t \in \{0, 1\}$, for example with $\mathbb{P}(x_t = 1) = 0.2$ and a fixed seed. To make the effect measurable, construct the target demand from the original baseline demand y_t^{base} as

$$y_t = y_t^{\text{base}} + \delta x_t, \qquad \delta > 0,$$

choosing δ to match the scale of the notebook (for example, $\delta = 8$ for unit sales). Do *not* add this adjustment if promotion already affects the original target.

Tasks to implement and analyze:

1. **Baseline model.** Reproduce FTL without a covariate: $\widehat{y}_t^{\text{base}} = \theta_t^{\text{base}}$, where θ_t^{base} minimizes the squared loss over past dates $s < t$. Set the initial forecast to 10.
2. **Enriched model.** For each t, calculate the counts $n_0(t)$ and $n_1(t)$, then θ_t and β_t *using only days* $s < t$. Apply the cold-start convention above. Produce $\widehat{y}_t^{\text{promo}} = \theta_t + \beta_t x_t$.

3. **Losses.** For the *same* observations y_t, calculate the two realized losses

$$L_T^{\text{base}} = \frac{1}{2}\sum_{t=1}^{T}(y_t - \widehat{y}_t^{\text{base}})^2, \qquad L_T^{\text{promo}} = \frac{1}{2}\sum_{t=1}^{T}(y_t - \widehat{y}_t^{\text{promo}})^2.$$

   Plot the evolution of both cumulative losses.

4. **Common comparator.** Calculate *after the fact* $w^\star = (\theta^\star, \beta^\star)$ that minimizes $\sum_{t=1}^{T}\widetilde{\ell}_t(w)$ over the entire period. Compare both strategies against this **same** comparator:

$$\text{Reg}_T^{\text{base|promo}} = L_T^{\text{base}} - \sum_{t=1}^{T}\widetilde{\ell}_t(w^\star),$$

$$\text{Reg}_T^{\text{promo}} = L_T^{\text{promo}} - \sum_{t=1}^{T}\widetilde{\ell}_t(w^\star).$$

   To plot **standard regret at each horizon** t, recompute the comparator $w_t^\star$ using only dates $s \le t$: $\text{Reg}_t = L_t - F_t(w_t^\star)$. The final curve at T therefore uses $w_T^\star$; also plot Reg_t/t. Use the same contextual comparator for the baseline strategy, so both models are compared on equal terms.

5. **Interpretation.** Plot y_t, both forecasts, promotion indicators, and the evolution of β_t. Compare errors on promotion days ($x_t = 1$) and non-promotion days ($x_t = 0$). Discuss the potential benefit of the covariate and what happens when promotions are rare.

Remark 2.5 (Implementation hints). At the beginning of each loop iteration, use x_t and history $\{(x_s, y_s) : s < t\}$. The variable y_t must be accessed **only after** the prediction has been recorded. Students may compute conditional means explicitly or check the result using `numpy.linalg.lstsq` once both categories have been observed. In general, fitting a model on all observations before the online loop would constitute information leakage.

Implementation timeline (pseudocode)

Inputs: demands $y_{1:T}$, indicators $x_{1:T}$, initial values $(\theta_1, \beta_1) = (10, 0)$.
For $t = 1, \dots, T$**:**

1. Estimate θ_t, β_t using only dates $s < t$.
2. Read x_t and record $\widehat{y}_t = \theta_t + \beta_t x_t$.
3. Reveal y_t and compute $\frac{1}{2}(y_t - \widehat{y}_t)^2$.
4. Add (x_t, y_t) to the history for round $t + 1$.

Outputs: forecasts, parameter trajectories, cumulative losses, and regrets.

Important note. The two regrets computed against the enriched comparator are not governed by the same theorem as regret for scalar FTL. Nevertheless, the constant model belongs to the enriched class ($\beta = 0$), which makes the operational comparison meaningful. An improvement in realized empirical loss must be demonstrated numerically: it is not guaranteed at every horizon.

3 Follow the Regularized Leader (FTRL)

3.1 Why regularize? A reminder of Ridge regression

In supervised statistical learning, observations $(X_i, Y_i)_{i=1}^n$ are used to fit a model $\mu_\beta(X_i)$. For linear regression with $\beta = (\beta_1, \dots, \beta_p)^\top$, Ridge regression minimizes the squared loss augmented by a penalty:

$$\widehat{\beta}^{\text{ridge}} \in \arg\min_{\beta_0 \in \mathbb{R},\, \beta \in \mathbb{R}^p} \left\{ \underbrace{\frac{1}{2}\sum_{i=1}^{n}(Y_i - \beta_0 - \beta^\top X_i)^2}_{\text{data fit}} + \underbrace{\frac{\lambda}{2}\|\beta\|_2^2}_{\text{penalty}} \right\}, \quad \lambda > 0.$$

The intercept β_0 is usually not penalized. As λ increases, explanatory coefficients are shrunk toward zero; as $\lambda \to 0^+$, we recover ordinary least squares when the limiting solution is well defined. An ℓ_2 penalty *does not generally make coefficients exactly zero*, unlike an ℓ_1 penalty such as Lasso.

3.2 Adding a constant does not change a minimizer

Example 3.1 (Adding a constant to the objective)**.** Consider the elementary problem

$$u^\star = \arg\min_{u\in\mathbb{R}} (u-1)^2 = 1.$$

If we add a constant independent of u, for example 170, then

$$u^\star = \arg\min_{u\in\mathbb{R}} \{(u-1)^2 + 170\} = 1.$$

Indeed, the derivative is unchanged and, more generally,

$$\arg\min_{u\in\mathcal{K}} \{f(u) + C\} = \arg\min_{u\in\mathcal{K}} f(u), \qquad C \in \mathbb{R}.$$

The *minimum value* increases by C, but the *minimizing point* is unchanged. This principle will allow us to remove terms that do not depend on u in linearized FTRL.

3.3 FTRL: stabilizing online decisions

FTL minimizes only the past losses. FTRL adds a regularizer R to control the magnitude or variation of the decisions.

Definition 3.2 (Follow the Regularized Leader)**.** Given $R : \mathcal{K} \to \mathbb{R}$ and $\eta > 0$, FTRL chooses

$$w_{t+1} \in \arg\min_{u\in\mathcal{K}} \left\{ \sum_{s=1}^{t} \ell_s(u) + \frac{1}{\eta} R(u) \right\}.$$

For example, we may use quadratic regularization: $R(u) = \frac{1}{2}\|u\|_2^2$, giving a term $\frac{1}{2\eta}\|u\|_2^2$.

Here η controls the *inverse* weight of regularization. Regularization increases the curvature of the cumulative objective and can prevent a small change in observations from producing a large shift in its minimizer.

3.4 What does it mean to "linearize" a function?

Recall the first-order Taylor expansion. If $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable at w, then, as $u \to w$,

$$f(u) = f(w) + \langle \nabla f(w), u - w \rangle + o(\|u - w\|_2).$$

The affine function

$$u \longmapsto f(w) + \langle \nabla f(w), u - w \rangle$$

is the **affine linearization** of f near w. In one dimension, this is the equation of the tangent line.

Example 3.3 (Linearizing a quadratic function)**.** For $f(u) = (u-1)^2$ and a point $w \in \mathbb{R}$,

$$f(u) = \underbrace{(w-1)^2 + 2(w-1)(u-w)}_{\text{affine part}} + \underbrace{(u-w)^2}_{o(|u-w|) \text{ as } u\to w}.$$

The quadratic term is negligible compared with $|u - w|$ near w. The affine approximation is therefore accurate *locally*.

Remark 3.4 (Differentiability and convexity). For a nondifferentiable convex function, the ordinary Taylor expansion cannot be invoked. However, if $g \in \partial f(w)$, then $f(w) + \langle g, u - w\rangle$ is a **global affine lower bound** on f, exact at $u = w$. This property justifies linearized FTRL based on subgradients.

3.5 Linearized FTRL: the connection with subgradients

At time s, choose a subgradient $g_s \in \partial \ell_s(w_s)$. The first-order inequality gives

$$\ell_s(u) \geq \ell_s(w_s) + \langle g_s, u - w_s\rangle.$$

The right-hand side is an **affine lower bound, exact at** w_s, but not necessarily a Taylor approximation in the usual sense when the function is nondifferentiable. We may use it as a *linearized surrogate loss*:

$$\widehat{\ell}_s(u) = \ell_s(w_s) + \langle g_s, u - w_s\rangle.$$

After substituting these losses into the regularized objective, the terms independent of u disappear from the argmin:

$$\begin{aligned} w_{t+1} &\in \arg\min_{u\in\mathcal{K}} \left\{ \sum_{s=1}^{t} \widehat{\ell}_s(u) + \frac{1}{2\eta}\|u\|_2^2 \right\} \\ &= \arg\min_{u\in\mathcal{K}} \left\{ \left\langle \sum_{s=1}^{t} g_s, u \right\rangle + \frac{1}{2\eta}\|u\|_2^2 \right\}. \end{aligned}$$

Proposition 3.5 (Explicit linearized FTRL update). *For $\mathcal{K} = \mathbb{R}^d$ and $R(u) = \frac{1}{2}\|u\|_2^2$,*

$$w_{t+1} = -\eta \sum_{s=1}^{t} g_s.$$

For a nonempty closed convex set $\mathcal{K}$, the solution is the Euclidean projection

$$w_{t+1} = \Pi_{\mathcal{K}} \left(-\eta \sum_{s=1}^{t} g_s \right).$$

Proof. By completing the square,

$$\left\langle \sum_{s\leq t} g_s, u \right\rangle + \frac{1}{2\eta}\|u\|_2^2 = \frac{1}{2\eta} \left\| u + \eta \sum_{s\leq t} g_s \right\|_2^2 - \frac{\eta}{2} \left\| \sum_{s\leq t} g_s \right\|_2^2.$$

The final term is constant in u. Without constraints, the minimizer is minus the cumulative gradient times η; with closed convex constraints, it is projected onto $\mathcal{K}$. □

Remark 3.6 (FTL, FTRL, and linearized FTRL). FTL minimizes the *actual* past losses. FTRL minimizes those losses with regularization. *Linearized* FTRL replaces each loss with its affine lower bound constructed at the decision actually played. These are different objectives, even though all three approaches follow the principle of sequential decisions.

3.6 Optional notebook extension: online Ridge regression

For the promotion model $w = (\theta, \beta)$, estimation may be stabilized by penalizing only the promotion coefficient:

$$(\theta_t, \beta_t) \in \arg\min_{\theta,\beta\in\mathbb{R}} \left\{ \frac{1}{2} \sum_{s=1}^{t-1} (y_s - \theta - \beta x_s)^2 + \frac{\lambda}{2}\beta^2 \right\}, \quad \lambda > 0.$$

This is a variant of FTRL (Ridge regularization on the promotion coefficient). For $t \geq 2$, it has a unique minimizer as soon as at least one past observation is available; in particular, $\beta_t = 0$ when only one category of x has been observed. As an additional exercise, compare the curves of β_t with

and without the penalty, for several values of λ, and discuss the trade-off between stability and adaptation.

4 Further exploration: contextual forecasting and temporal memory

In our first examples, the learner directly chose a number θ_t as the forecast:

$$\widehat{y}_t = \theta_t, \qquad \ell_t(\theta_t) = \frac{1}{2}(\theta_t - y_t)^2.$$

But a forecast may also depend on **features known when the decision is made**. The online decision then concerns the *parameters of a predictor*, rather than the predicted quantity itself.

4.1 Exogenous covariates and observed lags

Suppose that $x_t \in \{0, 1\}$ indicates whether a promotion is scheduled for day t. It is also natural to use recent observed sales $y_{t-1}, y_{t-2}, \ldots$. We might be tempted to define the vector $\boldsymbol{z}_t = (x_t, y_1, \ldots, y_{t-1})$. **The temporal idea is sound, but this vector grows with** t: its dimension changes, preventing direct use of a fixed-dimensional parameter $w \in \mathbb{R}^d$ in the usual FTL, FTRL, and OGD formulas.

For the notebook, let us therefore choose a **fixed** number of lags $p \geq 1$ (for example $p = 2$) and the vector

$$\boldsymbol{z}_t = (1,\, x_t,\, y_{t-1}, \ldots, y_{t-p})^\top \in \mathbb{R}^{p+2}, \qquad t \geq p + 1.$$

The first coefficient is an intercept, and $\boldsymbol{z}_t$ is available before y_t is revealed. The first p observations initialize the lags.

Definition 4.1 (Contextual linear prediction)**.** The online parameter is $w_t \in \mathcal{K} \subset \mathbb{R}^{p+2}$; the forecast and loss are

$$\widehat{y}_t = w_t^\top \boldsymbol{z}_t, \qquad \ell_t(w_t) = \frac{1}{2}(w_t^\top \boldsymbol{z}_t - y_t)^2.$$

The protocol is: observe $\boldsymbol{z}_t$, choose w_t using only past data, issue $\widehat{y}_t$, reveal y_t, record the loss, and then update w_{t+1}.

4.2 Three strategies for the same model

Let $t_0 = p + 1$ be the first evaluation day and $R(w) = \frac{1}{2}\|w\|_2^2$ be a possible regularizer. For $t \geq t_0$, the three strategies use the **same** observations and the **same** loss function:

$$\textbf{FTL:} \quad w_t \in \arg\min_{w \in \mathcal{K}} \sum_{s=t_0}^{t-1} \ell_s(w),$$

$$\textbf{FTRL:} \quad w_t \in \arg\min_{w \in \mathcal{K}} \left\{ \sum_{s=t_0}^{t-1} \ell_s(w) + \lambda R(w) \right\},$$

$$\textbf{OGD:} \quad w_{t+1} = \Pi_{\mathcal{K}}(w_t - \eta_t \nabla \ell_t(w_t)).$$

For OGD, the vector gradient is explicitly

$$\nabla \ell_t(w_t) = (w_t^\top \boldsymbol{z}_t - y_t)\boldsymbol{z}_t.$$

At date t_0, FTL has an empty loss history: we must specify a common initial choice w_{t_0} and a tie-breaking rule for nonunique minimizers (for example, the minimum-norm solution). For FTRL, $\lambda > 0$ penalizes the coefficient vector (potentially excluding the intercept, which must be stated explicitly).

Remark 4.2 (Dimension changes the geometry of the loss). The Hessian of a single day's loss is

$$\nabla^2 \ell_t(w) = \boldsymbol{z}_t \boldsymbol{z}_t^\top .$$

This matrix has rank at most one. As soon as $d = p + 2 > 1$, the individual loss is not strongly convex on $\mathbb{R}^d$. The *sum* of the losses may become strongly convex if the features are sufficiently varied, but the assumptions of the preceding logarithmic regret theorem are not automatically satisfied.

Important note. Using y_t, y_{t+1}, or lagged variables computed with future observations before issuing the forecast for day t is prohibited. Normalization parameters must not be fitted using future observations either. The same dates $t_0, \dots, T$ must be used to compare the three strategies.

Key takeaway. The initial model learns a constant forecast; the contextualized model learns $\widehat{y}_t = \theta_t + \beta_t x_t$. FTRL aims to stabilize this estimation. In both cases, *the forecast must always precede observation of* y_t.

Assignment due — November 15, 2026, 11:59 p.m.

Objective. Start from the illustrative notebook `Illustration_FTL_FTRL_OGD.ipynb` and apply **FTL, FTRL, and OGD** to the *same contextual model* with promotions and temporal memory. The interactive reference `circular_block_bootstrap_interactive.html` explains circular block bootstrap resampling.

1. Define the new feature set. The indicator $x_t \in \{0, 1\}$ represents a promotion on day t, known before the decision. Use a fixed number of lags, for example $p = 2$, and define for $t \geq p + 1$:

$$\boldsymbol{z}_t = (1, x_t, y_{t-1}, \ldots, y_{t-p})^\top, \quad \widehat{y}_t = w_t^\top \boldsymbol{z}_t.$$

For this assignment, retain p lags (or justify an equivalent fixed-dimensional representation) and move the window forward by one time step at each round.

2. Implement and compare the three algorithms. Follow the **forecast → observation → loss → update** protocol. Document all initializations and specify regularization coefficients and learning rates. On each day t, compute

$$\ell_t(w_t) = \frac{1}{2}(\widehat{y}_t - y_t)^2, \qquad L_t^A = \sum_{s=t_0}^{t} \ell_s(w_s^A), \quad A \in \{\text{FTL}, \text{FTRL}, \text{OGD}\}.$$

For a fair comparison, use **one common fixed comparator in hindsight** from the enriched linear model class:

$$w_t^\star \in \arg\min_{w \in \mathcal{K}} \sum_{s=t_0}^{t} \ell_s(w), \qquad \text{Reg}_t^A = L_t^A - \sum_{s=t_0}^{t} \ell_s(w_t^\star).$$

The denominator of the average-regret curve must be the number of *evaluated rounds*, $n_t = t - t_0 + 1$. Plot Reg_t^A / n_t. If you instead choose Reg_t^A / t, you must explain how the first p days are handled.

3. Plot forecasts and uncertainty.

- **Visualization:** plot the observed values y_t, the three forecasts $\widehat{y}_t^A$, and their empirical **95 % prediction intervals** on the same figure.
- **Circular block bootstrap (CBB):** resample consecutive blocks of length ℓ to preserve local temporal dependence. Fix a random seed, report the number of bootstrap replications B, and justify the choice of ℓ.
- **Sequential construction:** at time t, use only observations strictly earlier than t. Resample the available history, refit the predictor, then simulate an innovation from past forecast residuals, explicitly stating the assumptions made about these errors.
- **Prediction intervals:** compute the empirical 2.5 % and 97.5 % quantiles of the simulated predictions to obtain nominal 95 % prediction intervals.
- **Limitations:** these intervals are approximate and do not enjoy any universal coverage guarantee.

4. Uncertainty in average regret. For each algorithm, also plot the sequence $\left(\text{Reg}_t^A / n_t\right)_{t=t_0}^{T}$, together with an **empirical 95 % bootstrap interval**. For each replication $b = 1, \ldots, B$, resample the aligned observations (x_s, y_s) *using circular blocks*, reconstruct the lags $y_{s-j}^{*(b)}$ in the resampled series, **rerun FTL, FTRL, and OGD from scratch**, then recompute the comparator $w_t^{\star,(b)}$ and regret $\text{Reg}_t^{A,(b)}$. For each horizon t, define

$$\text{CI}_{95\%}^{\text{boot}}\left(\frac{\text{Reg}_t^A}{n_t}\right) = \left[Q_{0.025}\left(r_t^{A,*(1:B)}\right), Q_{0.975}\left(r_t^{A,*(1:B)}\right)\right],$$

where $r_t^{A,*(b)} = \text{Reg}_t^{A,(b)} / n_t$. These bounds describe *empirical resampling variability*; they must not be presented as regret guarantees valid for every adversarial sequence.

5. Required analysis. Discuss performance on promotion and non-promotion days, the effect of the p lags, the OGD learning rate, FTRL regularization, the block length ℓ, and the distinction between *realized regret*, *prediction intervals*, and *bootstrap intervals for average regret*. State the limitations of the bootstrap procedure (temporal dependence, local stationarity, boundary effects between blocks, and small sample sizes).

Deliverable. One fully executable `.ipynb` notebook containing completed code cells, labeled figures, a fixed random seed, documented parameters, and written Markdown answers. **Deadline: Sunday, November 15, 2026, at 11:59 p.m.**